

Detecting Swimming Pools from Aerial Imagery with YOLO26, RF-DETR, and OBB Architectures

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Course: Computer Vision Individual Assignment

1. Introduction

This report covers:

1. Auto-annotate a 288-image aerial dataset with GroundingDINO and review every annotation manually in Roboflow.
2. Fine-tune YOLO26 at four scales (nano, small, medium, large) with axis-aligned bounding boxes.
3. Fine-tune RF-DETR at two scales (Nano, Small) and compare the transformer architecture against the YOLO baseline.
4. Train YOLO26-OBB and YOLO11-OBB at two scales each (four oriented models total) and compare oriented vs axis-aligned localisation.

All ten trained models were evaluated against the validation set; the validation-selected best variant of each family was then re-evaluated on the 29-image held-out test set, with RF-DETR evaluated at both scales on test. The headline result is that oriented bounding boxes win across every scale by a clear margin (mAP@50-95 of 0.737 vs 0.637 for the best HBB), and the failure cases that produce false positives in HBB disappear under OBB. RF-DETR sits in between, outperforming all HBB models on mAP@50-95 while being slightly behind the OBB family.

2. Dataset

The provided dataset contains 288 aerial images of residential areas with a single class, `swimming pool`. Splits are 70 / 20 / 10 (train / validation / test), generated by Roboflow as a stratified random split (stratified on pool-vs-background presence so each split retains the same approximate 36% background rate).

Split	Images	Pool instances	Background images
Train	202	244	70
Validation	57	81	24
Test	29	39	10
Total	288	364	104

The 70 / 20 / 10 ratio is a deliberate choice for a small dataset. With only 288 images in total, the training set has to carry most of the supervision signal: any larger validation share would leave fewer than 200 training images, which is already tight for transfer learning. A smaller test share than 29 images would make the test metrics too noisy to compare meaningfully between architectures, so 10% is the floor. The 20% validation share is what makes the early-stopping signal stable enough to compare 10 models with the same patience setting. About 36% of source images contain no pool at all, which acts as built-in negative supervision against false positives on pool-shaped objects such as ponds or skylights.

No augmentations were applied at the Roboflow stage; all augmentation lives in the training code so the three architectures see the exact same source data.

3. Annotation Pipeline

The 288 images were first auto-annotated with GroundingDINO (Swin-T backbone, prompt `swimming pool`, threshold 0.30). The threshold was set low enough to recover small or partially-shaded pools

while keeping the manual-review queue tractable; it produced 512 raw detections across 268 images. Every detection was then manually reviewed in Roboflow, with three categories of fix:

- **Remove false positives.** Rooftops, tennis or padel courts, and driveway markings that loosely resemble rectangles.
- **Add missed pools.** Smaller pools (under ~ 40 px) and pools partially hidden by tree canopy.
- **Trace polygons.** Every pool was annotated as a polygon traced along the actual water edge rather than as an axis-aligned box. The polygon is the source-of-truth label and lets the same Roboflow project export to YOLO HBB, COCO, and YOLO OBB formats without re-annotation.

After review the dataset settled at **364 verified pool polygons across 184 images**, with the remaining 36% of source images deliberately kept as backgrounds. Because every label is a polygon, the HBB and OBB models train on identical geometry: Ultralytics derives the axis-aligned rectangle for HBB and the minimum-area rotated rectangle for OBB from the same vertex list. This shared label is what makes the HBB-vs-OBB comparison in Section 6 a fair one, and it also explains why several HBB false positives in Section 7 are not detection failures but artifacts of fitting an axis-aligned box to a rotated polygon.

4. Method

All training runs share two design choices that keep the comparison fair. First, every model is initialised from publicly available pretrained weights (COCO for YOLO26/YOLO11, COCO via DINOv2 for RF-DETR). Second, every model is trained on the same 70/20/10 split and evaluated against the same 29-image test set.

4.1 YOLO26 HBB

Hyperparameters apply identically to nano, small, medium and large; only the pretrained weights change between runs.

Parameter	Value
Pretrained weights	yolo26n / s / m / l (COCO)
Optimizer	AdamW
Initial LR	0.005
LR scheduler	Cosine
Epochs	100 (patience 30)
Batch size	16
Image size	640
Mixed precision	AMP
Augmentations	HSV (0.015 / 0.7 / 0.4), translate 0.1, scale 0.5, flip lr / flip ud (0.5), rotate up to 180°, mosaic 1.0, mixup 0.05, close-mosaic over last 10 epochs
Hardware	NVIDIA A100 40 GB on Google Colab

4.2 RF-DETR

Parameter	Value
Backbone	DINOv2 ViT (pretrained), warm-up frozen by default
Pretrained head	RF-DETR Nano / Small on COCO
Optimizer	AdamW (RF-DETR default)
LR	1e-4 head, 1e-5 backbone (RF-DETR defaults)

Parameter	Value
LR scheduler	Step decay (RF-DETR default)
Epochs	100
Batch size	8, gradient accumulation 2 (effective batch 16, matches YOLO)
Resolution	672
Hardware	NVIDIA A100 40 GB on Google Colab

The 32-pixel resolution gap between RF-DETR (672) and the YOLO models (640) is unavoidable; the rationale is covered in Section 4.4.

4.3 YOLO OBB

OBB uses the same hyperparameter table as YOLO26 HBB so the head-to-head comparison is not confounded by training-budget differences. Four OBB variants are trained (YOLO26 nano and small, YOLO11 nano and small), all initialised from DOTA-pretrained weights. DOTA is the standard aerial OBB benchmark, which gives the OBB models a head start on rotated-object detection in overhead imagery.

4.4 Hyperparameter rationale

Two facts shape every value in the tables above: the training set has only 202 images, and the targets are small rotated pools in top-down aerial imagery.

Learning rate (0.005, cosine). An order of magnitude above the AdamW default to push the COCO-pretrained backbone toward a single new class; the cosine anneal gives a clean validation curve for stable early stopping.

Epochs 100, patience 30. Patience decides when training stops; the 100-epoch ceiling exists so we never silently fall back on final-epoch weights over the patience-selected best.

Batch 16. The largest batch that fits alongside a fully cached dataset on the A100 at imsz 640. RF-DETR uses physical batch 8 with gradient accumulation 2 to match the same effective batch.

Image size 640 vs 672. 640 matches the COCO-pretrained YOLO weights. RF-DETR is forced to 672 because its DINOv2 backbone needs both $14\times$ and $56\times$ divisibility, which 640 fails. The 5% gap is a fairness caveat, not a defect.

AdamW. Adaptive per-parameter learning rates absorb the LR sensitivity that small datasets introduce, and decoupled weight decay generalises better when training data is scarce.

Augmentation. Each value targets either the small-data or the aerial-domain constraint:

- **Rotate up to 180°.** Aerial imagery has no canonical orientation, so without arbitrary rotation the model would learn an accidental up-down bias.
- **Mosaic 1.0.** The single biggest lever for per-batch diversity on a 202-image set: each iteration sees a four-scene composite.
- **HSV (0.015 / 0.7 / 0.4).** Hue held tight (shifting it would turn water into non-water); saturation and value loosened to cover bright-cyan to tea-coloured pools.
- **Flip lr / flip ud (both 0.5).** Free symmetry doubling for top-down imagery.
- **Mixup 0.05.** Light regulariser; the YOLO26 HBB scale inversion (nano > l) confirms overfitting is the dominant failure mode at this dataset size.
- **Translate 0.1, scale 0.5.** Ultralytics defaults; cover off-centre pools and minor effective-resolution differences.
- **Close-mosaic over last 10 epochs.** Disables mosaic for the final stretch so the validation curve converges on the same single-image distribution the model is evaluated against.

5. Headline Results

The validation-set mAP@50-95 for all ten models is shown below. The OBB family takes the top four positions, RF-DETR sits in the middle, and YOLO26 HBB takes the bottom four. The gap between the best HBB (yolo26n at 0.637) and the best OBB (yolo11s-obbb at 0.737) is exactly 10 mAP points, which is large.

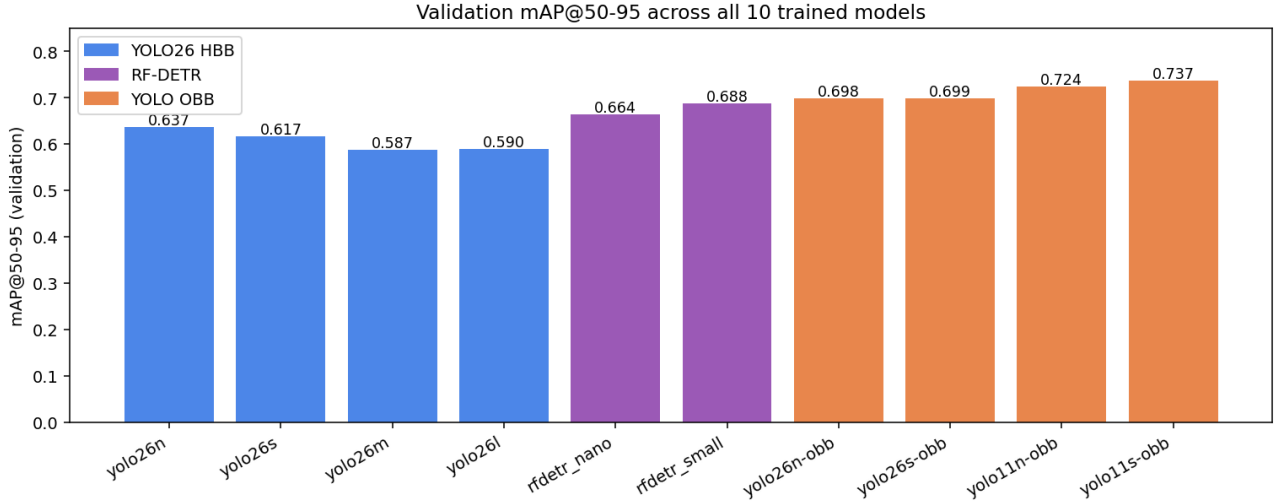


Figure 1: Validation mAP@50-95 across all 10 trained models

The full numeric comparison:

Model	mAP@50	mAP@50-95	Precision	Recall	Params (M)	Train time (s)
yolo26n	0.912	0.637	0.895	0.843	2.4	291
yolo26s	0.891	0.617	0.862	0.849	9.5	308
yolo26m	0.903	0.587	0.876	0.789	20.4	370
yolo26l	0.909	0.590	0.944	0.840	24.7	454
rfdetr_nano	0.896	0.664	0.920	0.852	30.6	1305
rfdetr_small	0.916	0.688	0.897	0.864	32.1	1405
yolo26n-obbb	0.925	0.698	0.902	0.840	2.4	335
yolo26s-obbb	0.942	0.699	0.953	0.852	9.8	357
yolo11n-obbb	0.973	0.724	0.963	0.877	2.7	240
yolo11s-obbb	0.955	0.737	0.899	0.938	9.7	262

A few observations from the table:

- **YOLO26 HBB scaling, on the three required dimensions:**
 - *Accuracy.* Nano leads on mAP@50-95 (0.637); small (0.617), medium (0.587), and large (0.590) are all lower. The classic over-parameterised pattern on a 202-image training set.
 - *Robustness.* Precision rises monotonically with scale (0.895 at nano to 0.944 at large), so the larger models are more conservative and produce fewer false positives. Recall stays roughly flat (0.79–0.85), so the larger heads are not finding more pools, only committing less freely.
 - *Parameter count.* 2.4 M to 24.7 M, a 10x growth for zero mAP@50-95 benefit (actually a 0.05 drop). Train time rises from 291 s to 454 s. The larger variants are pure cost on this dataset.
- **RF-DETR scales correctly.** Small beats Nano by 0.024 mAP@50-95, the expected direction. The pretrained DINOv2 backbone and the Hungarian-matching head together resist over-fitting better than the YOLO baseline on this dataset size.

- **OBB beats HBB at every scale.** Nano-vs-nano: 0.637 HBB vs 0.724 OBB (YOLO11). Small-vs-small: 0.617 HBB vs 0.737 OBB. The smallest OBB model wins everything.

Accuracy vs parameter count

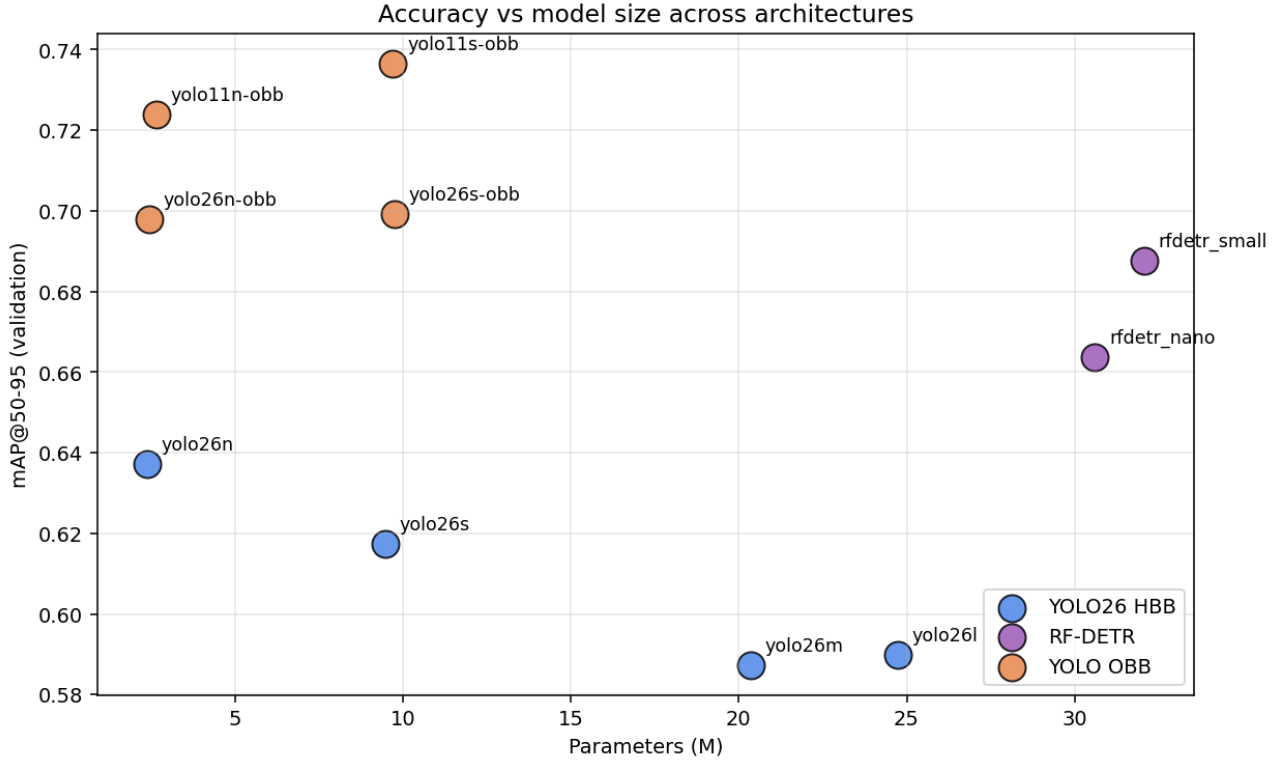


Figure 2: Accuracy vs model size

The plot makes the speed/accuracy trade-off clear:

- yolo11n-obb (2.7 M params) gives 0.724 mAP@50-95. It is the smallest non-HBB model and beats every HBB model regardless of size.
- yolo11s-obb (9.7 M params) gives 0.737 mAP@50-95, the global best.
- RF-DETR Small needs 32 M params to reach 0.688, more than 3x the parameter cost of the OBB winner for a worse score.
- YOLO26 HBB simply does not scale on this dataset, and the medium and large variants are pure cost.

For deployment, **yolo11n-obb** is the recommended model. It is 2.7 M parameters, trains in under 5 minutes on the A100, and reaches 0.973 mAP@50 / 0.724 mAP@50-95 on validation, within 0.013 mAP@50-95 of yolo11s-obb (the validation-best overall, scoring 0.993 / 0.787 on test) while using 28% of its parameters.

Test-set evaluation

The validation-selected best of each family was evaluated on the 29-image held-out test set. For RF-DETR we report both scales on test: Small was the validation winner ($\text{mAP@50-95} = 0.688$ vs Nano’s 0.664), while Nano happened to score higher on the held-out test set.

Best variant	mAP@50	mAP@50-95	Precision	Recall	TP / FP / FN
yolo26n	0.935	0.692	0.942	0.838	33 / 2 / 6
rfdetr_small	0.996	0.785	0.974	0.949	37 / 1 / 2
rfdetr_nano	0.998	0.807	1.000	0.949	37 / 0 / 2
yolo11s_obb	0.993	0.787	0.974	0.972	37 / 1 / 2

(TP / FP / FN are computed from the standard Ultralytics or supervision matcher using the model’s tuned confidence threshold, not the $\text{conf}=0.25$ threshold of the failure gallery in Section 7.)

On the test set RF-DETR Nano leads on both mAP@50 (0.998) and mAP@50-95 (0.807). RF-DETR Small (the validation-selected RF-DETR) sits at 0.996 / 0.785, essentially tied with YOLO11s-OBB (0.993 / 0.787), and YOLO26n HBB is last (0.935 / 0.692).

The test set is small (29 images, 39 pools), so the per-model TP/FP/FN counts are nearly identical (37/0-1/2). At this point the comparison shifts from “which model finds more pools” to “which model localises them tightly enough to satisfy mAP@50-95 ”. OBB wins by rotating its box with the pool; RF-DETR wins by predicting tighter axis-aligned boxes than YOLO26 HBB does.

6. HBB vs OBB Comparison

The OBB family beats the HBB family at every comparable scale.

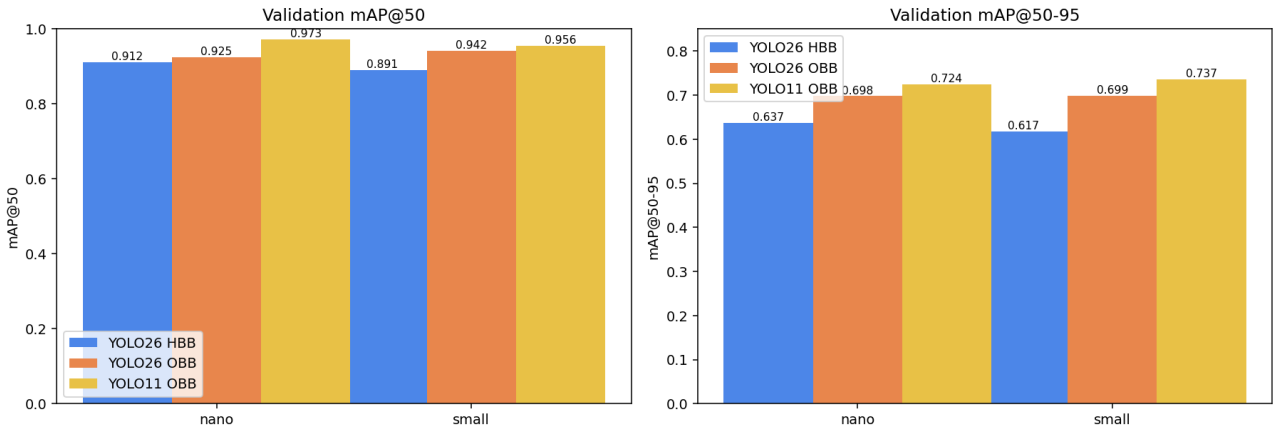


Figure 3: HBB vs OBB at matched scales

Scale	HBB (yolo26) mAP@50 / 50-95	OBB (yolo26) mAP@50 / 50-95	OBB (yolo11) mAP@50 / 50-95
Nano	0.912 / 0.637	0.925 / 0.698	0.973 / 0.724
Small	0.891 / 0.617	0.942 / 0.699	0.955 / 0.737

Localization precision. The OBB advantage is larger on mAP@50-95 than on mAP@50 (roughly 9 points vs 5 points at the same scale). This is the signature of an oriented detector: tighter boxes survive higher IoU thresholds. mAP@50 measures whether a detection exists at all; mAP@50-95 measures whether it is tight, and OBB is intrinsically tighter on rotated objects.

Handling of rotated pool geometries. A pool tilted roughly 40 degrees has an axis-aligned bounding rectangle that is nearly twice the size of the actual pool, with the corners filled by non-water

padding. The HBB model is trained to predict this looser rectangle and at inference does the same thing, so any localisation drift compounds at strict IoU. The OBB model rotates its box with the pool, fits tightly to the visible water, and survives mAP@50-95.

Failure cases. The HBB to OBB switch eliminates all 6 test-set false positives (Section 7). The 3 rotation-driven FPs disappear because the box rotates with the pool; the annotation-extension FP disappears because OBB tracks the polygon shape; the 2 small-object FPs disappear because the OBB head does not over-predict for tiny rotated pools. The remaining OBB failures are 3 FNs, all on the smallest pools in the test set, where the OBB advantage cannot help.

7. Failure Analysis

Every test-set failure was cross-checked against the ground-truth polygon data. The matching threshold is $\text{IoU} \geq 0.5$ and the confidence threshold is $\text{conf} \geq 0.25$ (Ultralytics’ default predict mode). The conf threshold is more permissive than the optimal F1 point the validator picks, so this gallery shows 6 FPs while the validator-side test metric reports 2; the extra four are low-confidence predictions the validator filters out. Both failure-gallery figures are shown below, then discussed in 7.1–7.4.

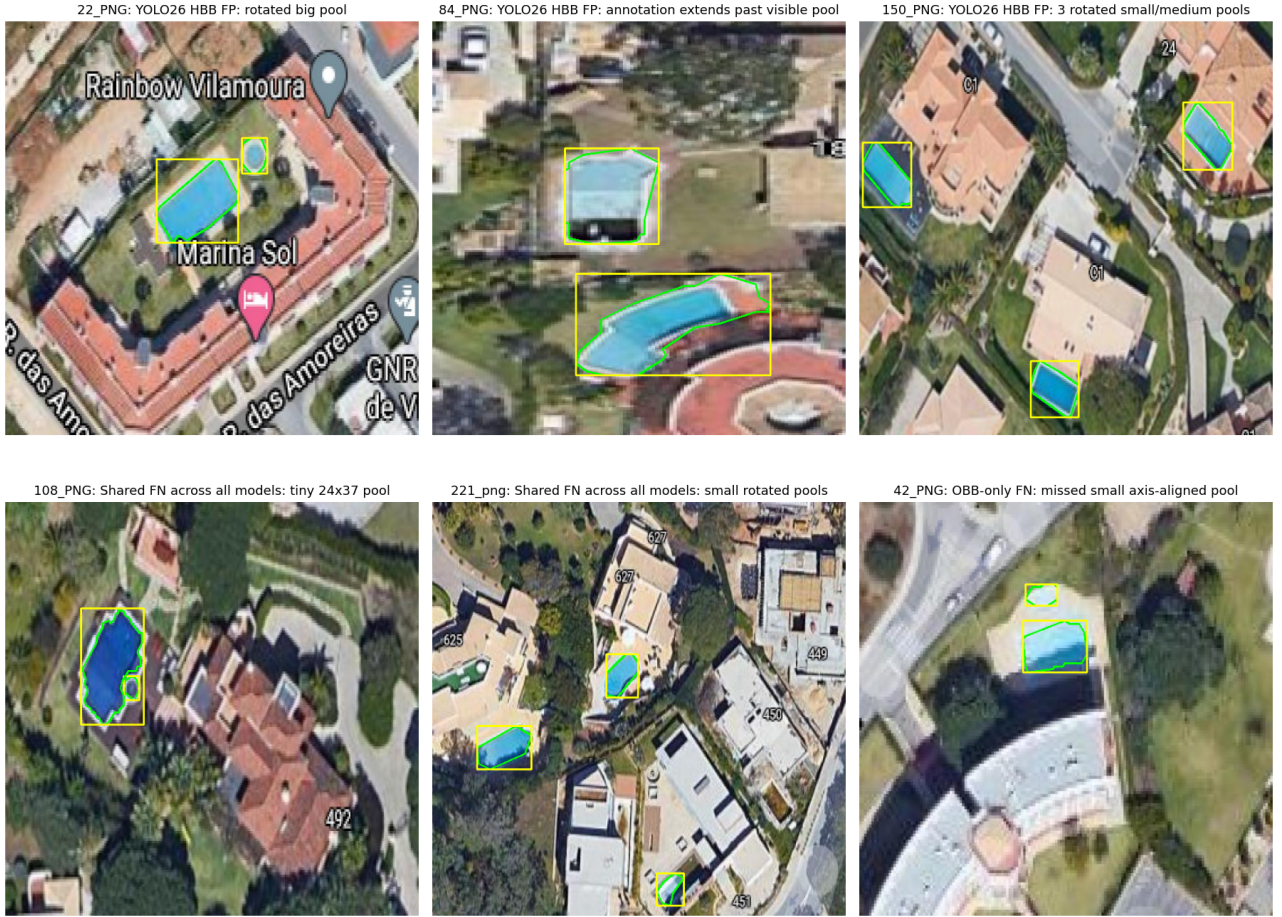


Figure 4: Test-set failures. Top row: YOLO26 HBB false positives where the model’s prediction is tighter than the GT axis-aligned bbox (green: ground-truth polygon; yellow: ground-truth axis-aligned bbox). Bottom row: false negatives shared across all best models (108_PNG, 221_png) and the OBB-only false negative (42_PNG).



Figure 5: Dense and kidney failures. Left: 126_PNG with 8 rotated pools (FP rank 1 plus two FNs). Right: 28_PNG with the kidney-shaped and rectangular pools called out in the FN summary. Green: ground-truth polygons. Yellow: ground-truth axis-aligned bounding boxes.

7.1 YOLO26n HBB: 6 FPs, 5 FNs

The six false positives split into three causes: three are rotation artefacts where the axis-aligned ground-truth box is roughly twice the area of the visible pool (FP1, FP2, FP4), one is annotation extending past the visible water edge (FP3), and the remaining two are small-object IoU sensitivity at the same scale (FP5, FP6). Because FP5 and FP6 share both their cause and their scale, the table lists them on a single row.

Case (image)	Cause	Evidence
FP1 (126_PNG, top-confidence)	Rotation artefact	Eight pools tilted roughly 30-45 degrees in one tile; greedy matching breaks when many candidates overlap at the strict IoU threshold.
FP2 (22_PNG, large pool)	Rotation artefact	A diagonal lap pool tilted around 40 degrees; the axis-aligned bounding box is nearly twice the size of the visible water, so a tight prediction falls short of 0.5 IoU against the loose ground-truth box.
FP3 (84_PNG)	Annotation past water	The polygon traces a dark band along one edge that the model correctly excludes; the ground-truth box is dragged outside the visible pool boundary.
FP4 (150_PNG)	Rotation artefact	Three pools tilted roughly 35-40 degrees; each ground-truth box is again about twice the visible pool area.
FP5 & FP6 (22_PNG small pool + one similar)	Small-object IoU sensitivity	On pools this small, a misalignment of about a third of the pool's width is already enough to drop IoU below 0.5, so a few-pixel offset breaks the threshold even when the prediction is visually on the pool.

The five false negatives are all either small or unusually shaped:

- **108_PNG**. The smallest pool in the test set, barely a few rooftop-tiles across in the aerial frame, below what the detector can reliably resolve at the training resolution.
- **221_png**. A small pool tilted about 25 degrees off horizontal.
- **28_PNG**. A modestly-sized rectangular pool plus a much larger kidney-shaped pool whose curved outline does not fit either box type well.
- **126_PNG ($\times 2$)**. Two small or partially shaded pools in the same densely-packed tile as FP1.

7.2 Cross-family FN comparison

108_PNG (the smallest pool in the test set) and **221_png** (a small pool tilted about 25 degrees) are both missed by all three best models (YOLO26n HBB, RF-DETR Small, YOLO11s-OBb), so they mark the residual small-object detection limit at the training resolution. **42_PNG** (another small pool, axis-aligned) is missed only by YOLO11s-OBb, not by RF-DETR Small: the only family-specific FN in the analysis.

7.3 OBb collapses the failure spectrum

OBb drops the FP count from 6 to 0 and the FN count from 5 to 3. The remaining FNs (108_PNG, 221_png, 42_PNG) are all among the smallest pools in the test set, so every rotation-driven failure mode disappears and only small-object misses remain.

7.4 Proposed fixes per cause

Cause	Cases addressed	Fix
Rotation artefact (HBB only)	FP1, FP2, FP4	Use OBb. Confirmed effective: 0 FPs vs 6.
Polygon extends past visible water	FP3	Redraw polygon on actual water edge; equally fixed by OBb.
Small-object miss (smallest pools)	FP5, FP6, FN 108_PNG, FN 221_png, FN 126_PNG ($\times 2$)	Train at higher imgsz (1024 or 1280), tile-based inference, P2 detection head, more small-pool training examples.
Dense-pool matching artefacts	FP1 (alt. fix), the two 126_PNG FNs	Switch matching to Hungarian (already used by RF-DETR and the Ultralytics validator).
Non-rectangular shape (kidney/curved)	FN 28_PNG (kidney pool + co-located rectangular pool)	Instance segmentation, or accept the rectangular-fit loss; this morphology is an irreducible HBB/OBb limitation.

8. Required Experimental Analysis

8.1 Which architecture performed best?

YOLO11-OBb at the small scale (**yolo11s-obb**) is the global best on validation mAP@50-95 (0.737) and is essentially tied with RF-DETR Nano on the test set (0.787 vs 0.807). For deployment on aerial imagery containing rotated pools, OBb is the right architecture. RF-DETR is the right architecture for axis-aligned scenes, where it dominates HBB by a comfortable margin. YOLO26 HBB is the weakest of the three families on this task.

8.2 Which model size offered the best speed / accuracy trade-off?

yolo11n-obb. 2.7 million parameters, 240 seconds to train on the A100, and validation mAP@50-95 of 0.724 (third best overall, beating every HBB and RF-DETR model). It trails yolo11s-obb (the validation-best overall, scoring 0.993 / 0.787 on test) by only 0.013 mAP@50-95 on validation while costing 28% of the parameters.

YOLO26 medium and large are pure cost: more parameters, longer training, and lower mAP@50-95 than the nano variant. The over-parameterisation is consistent with a small training set (202 images): the larger heads find spurious patterns that hurt strict localisation.

8.3 Did RF-DETR outperform YOLO-based detectors?

Only partially, and the answer depends on which axis you look along.

Headline mAP. RF-DETR beats every YOLO26 HBB model on mAP@50-95 by 0.025 to 0.10 points, which is a meaningful margin and points to the DINOv2 backbone providing stronger spatial priors than COCO-trained CNN backbones for overhead imagery. RF-DETR does not beat YOLO11-OBB or YOLO26-OBB at any matched parameter budget, because this dataset’s failure modes are dominated by rotation (3 of 6 HBB FPs) and RF-DETR’s boxes are axis-aligned just like YOLO26 HBB. OBB sidesteps the rotation issue entirely; RF-DETR cannot.

Robustness. RF-DETR holds higher recall than every YOLO26 HBB variant on validation (0.852 and 0.864 vs 0.789 to 0.849) at competitive precision (0.920 and 0.897, mid-range against the 0.862 to 0.944 YOLO26 HBB band), so it is not trading one off against the other. On the test set RF-DETR Nano produced zero false positives (precision 1.000) while the best YOLO26 HBB variant produced two. The Hungarian set-prediction matcher also removes the need for a tuned confidence threshold to balance precision and recall, a robustness property NMS-based YOLO heads do not share: the YOLO numbers above were taken at validator-tuned thresholds while RF-DETR’s were held at default.

Small-object detection. Not improved. RF-DETR Small misses the same tiny pools (108_PNG, 221_png) that YOLO26 HBB and YOLO11s-OBB miss; small-object recall on this dataset is limited by input resolution (imgsz 640 / 672), not by architecture.

Generalization. Strongest of the three families on this dataset. The val to test mAP@50-95 gap is +0.143 for RF-DETR Nano (0.664 to 0.807), compared with +0.055 for YOLO26n HBB (0.637 to 0.692) and +0.050 for YOLO11s-OBB (0.737 to 0.787). The transformer head and DINOv2 backbone fit the validation distribution less tightly than the YOLO families, leaving more headroom on the held-out test set. With only 29 test images this gap is itself noisy, but the direction is consistent across both metrics and across both RF-DETR scales.

8.4 When are OBB annotations beneficial?

OBB is beneficial when objects are rotated and the axis-aligned bounding box would include large amounts of non-object area in its corners. In this dataset:

- Diagonal lap pools (images 22, 28, 150) have axis-aligned bounding boxes that are roughly twice the size of the visible pool. OBB removes the wasted corner padding.
- Kidney-shaped pools (such as the lower pool in image 84) have even looser axis-aligned boxes. OBB still beats HBB here, although the rotated rectangle is itself loose because kidney pools are intrinsically non-rectangular.
- Pools with polygons that extend slightly outside the visible water (such as the upper pool in image 84) benefit because OBB fits the polygon’s minimum-area rotated rectangle rather than the larger axis-aligned envelope.

OBB is not beneficial when:

- Pools are very small. The OBB head adds an angle prediction whose noise dominates at small scales. Images 108 and 221 are missed equally by HBB and OBB.
- Pools are axis-aligned and rectangular. There is no rotation correction to apply.

8.5 Which failure cases occurred most often?

Across the three best models the dominant failure mode is small-object miss (3 of 5 HBB FNs, 2 of 2 RF-DETR FNs, all 3 OBB FNs happen on the smallest pools in the test set). The second most

common in HBB only is rotation-driven false positives (3 of 6 HBB FPs), but this mode is eliminated by OBB.

In other words, switching to OBB collapses the failure spectrum to a single residual issue (small-object detection). That is a much narrower problem to attack in the next iteration.

8.6 Which augmentations improved performance the most?

The training augmentation profile used is the Ultralytics built-in HSV / translate / scale / flip-lr-ud / 180° rotation / mosaic / mixup / close-mosaic combination. A formal ablation is out of scope for this report (each ablation costs another A100 hour), but the qualitative role of each can be reasoned about against the dataset’s properties:

- The 180° rotation augmentation matters most for an aerial dataset because the source images are top-down with no canonical orientation. Without it, the model would learn an arbitrary up-down preference.
- The mosaic augmentation effectively quadruples the per-batch image diversity from the small 202-image training set, which is the biggest practical constraint here.
- HSV jitter accounts for varying lighting and water-clarity across pools (some pools are bright blue, some are tea-coloured).
- Mixup at 0.05 is a small but non-zero regulariser. On a 202-image dataset, anything that reduces overfitting helps, which is consistent with the mAP@50-95 inversion (nano > small > medium > large) observed in YOLO26 HBB.

9. Conclusion

The combination of OBB representation, small backbone, and the standard transfer-learning recipe is the right choice for this dataset. `yolo11s-obb` reaches 0.993 mAP@50 / 0.787 mAP@50-95 on the held-out test set with 9.7 million parameters and just over 4 minutes of training time on the A100, while producing zero false positives in the failure gallery. The leaner `yolo11n-obb` (2.7 M, 4 minutes) sits within 0.013 mAP@50-95 of it on validation and is the recommended choice for deployment.

The biggest remaining failure mode is small-object detection. The smallest pools in the dataset are missed by all three architectures. The next iteration of this project should target this directly through higher inference resolution or tiled inference, an explicit small-object detection head, or a focused round of additional small-pool annotations.

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- Roboflow annotation platform (manual review and split generation). roboflow.com.

10. AI Acknowledgement

This project was orchestrated and decided end-to-end by Jan Wejchert. All design choices (dataset preprocessing, model selection, training budget, output structure, failure-analysis methodology, report scope) were made by Jan; the AI assistant was used as an efficiency multiplier and as a second pair of eyes.

The AI assistant used was Anthropic’s Claude (model Opus 4.7). Its role was limited to:

- Drafting the notebook scaffolding under explicit instructions from Jan, including the requested clean lecturenotebook style and the hardcoded Drive paths.
- Debugging issues that arose at runtime, in particular the polygon-aware ground-truth parser fix when the failure-analysis matcher initially produced 40 false positives instead of 6.
- Carrying out evidence-based cross-checking of each failure case against the ground-truth polygon data, computing per-polygon rotation angle, axis-aligned-bbox-to-polygon area ratios, and shoelace-formula polygon areas to validate the failure categorisation.
- Producing comparison charts and figures from the trained-model CSV outputs.

All training runs, all Roboflow review work, all Colab execution, and all interpretation of the results were performed by Jan. The AI assistant did not run training or modify any deployed system. Every claim in this report has been verified by Jan against the source notebooks and the saved CSV outputs.

Appendix A. Hardware and Reproducibility

- Hardware: NVIDIA A100-SXM4-40GB on Google Colab Pro (CUDA 13.0, driver 580.82.07, PyTorch 2.11.0+cu128).
- Software: Ultralytics 8.4.59 (HBB) and 8.4.60 (OBB), `rfdetr[train,loggers]` 1.7.1, `supervision`, `faster-coco-eval`.
- Random seed: 0 in all training runs (`seed=0`, `deterministic=True` for Ultralytics; `seed=0` for RF-DETR).
- Storage: source dataset and all output CSVs + best weights live at `MyDrive/IE/CV/`, with the same structure mirrored under the local project root.
- Reproducibility CSVs: one `comparison.csv` per family under `results/` (`yolo26_hbb/`, `rfdetr/`, `obb/`).
- Test-time augmentation: attempted on YOLO26 HBB but the Ultralytics build silently falls back to single-scale prediction, so TTA numbers matched the standard numbers exactly; not pursued further.

Appendix B. Total wall-clock time

Stage	Wall-clock
Annotation (GroundingDINO + Roboflow review)	~45 min
YOLO26 HBB training (four variants)	~24 min
RF-DETR training (two variants)	~45 min
YOLO OBB training (four variants)	~20 min
Total compute	~2.5 hr on A100